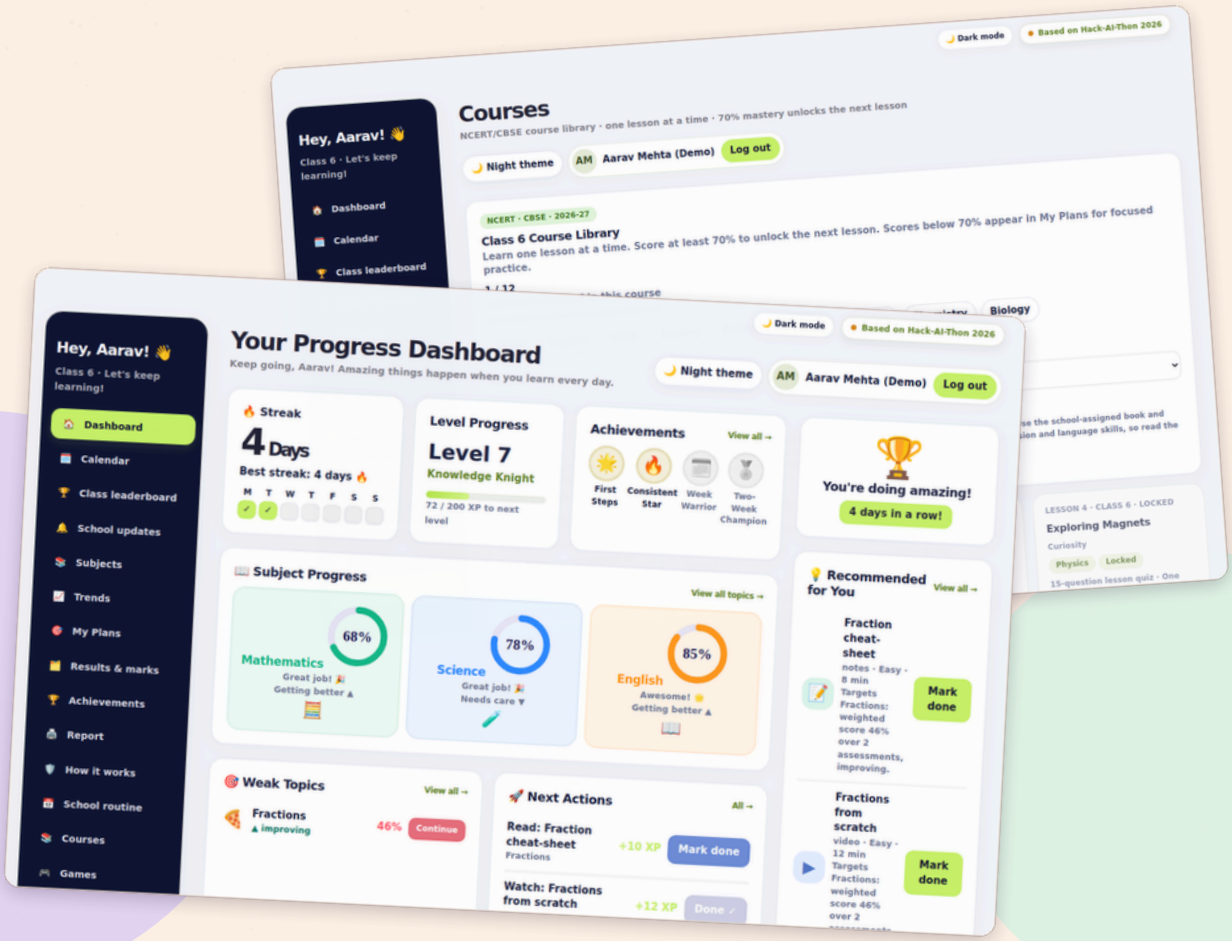
MYRA GLOBAL TECH HACK-AI-THON 2026

Learn4All

Inclusive Learning Progress & Resource Planner

Amastery-based learning companionfor Classes6to 10.
Students practise, plan and track progress; teachers seethewhole class at a glance. Built with plain Python and SQLite.



LEARN · PRACTISE · GROW · ACHIEVE

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How to read this document

Sections1 to 3explain the idea. Sections4 to 8 walk through each screen of the running application. Sections 9 to 13 cover design, technology, setup and what comes next.

About the screenshots

Every screenshot was captured from the running application using the built-in fictional demo data. Names carry a (Demo) label and no real learner is shown.

THE PROJECT IN ONE MINUTE

1 Executive summary

Studentslearn.Teacherssee.Learn4All turnsonesyllabus-linked lesson library into a calm, step-by-step practice loop.

Learn4All is an inclusive learning progress and resource planner for school students in Classes 6 to 10. A student signs in, picks a lesson from the NCERT-linked course library, takes a short multiple-choice quiz and needs **70%** to unlock the next lesson. Anything below the line lands in a personal revision list with a one-click retake. Every score feeds a dashboard, trend charts, achievements and a class leaderboard.

A teacher or school admin signs in to a separate workspace with a class overview heatmap, a marks updater, a monitoring list of students who need a check-in, and a notice board that can target everyone, one class or a single student. The whole application runs locally on plain Python with an SQLite database, so there is nothing to install beyond Python itself.

303 catalogue	**6-10** classes supported	**9** subjects and domains	**15** uniqueMCQs per quiz
3 levels: easy, medium, hard	**70%** masteryto unlock the next lesson	**7** explainable smart rules	**32** automated tests passing

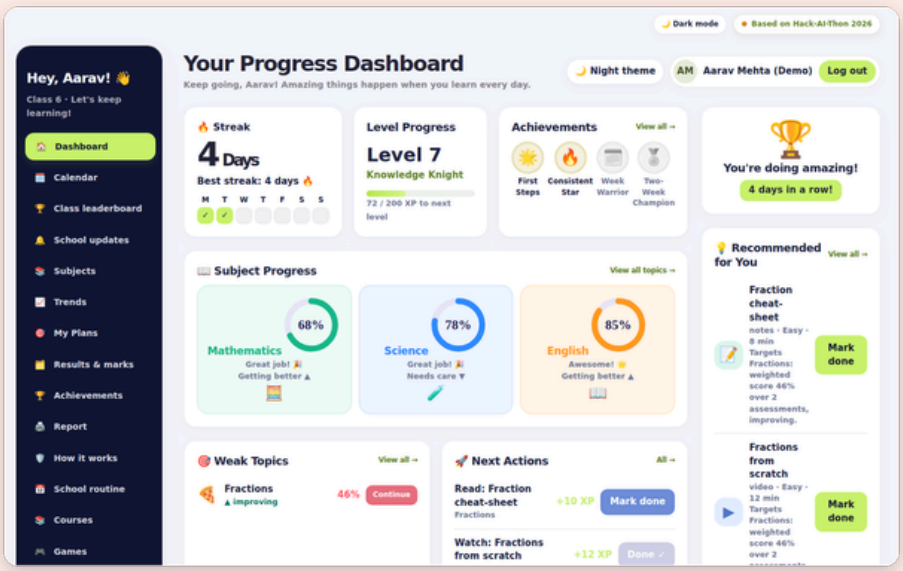


Figure 1. The student dashboard: streak, level, subject progress rings, weak topics and next actions.

WHY LEARN4ALL EXISTS

2 Problem and solution

Learners need a clear nextstep.Teachersneedtoknowwho needs help without reading every answer sheet.

Classroom learning often leaves students unsure what to study next, and leaves teachers with marks that arrive too late to act on. Learners also differ widely in how they read, hear and focus, so a single fixed format helps some and shuts out others. Learn4All responds to these needs with a small set of focused ideas:

Learner or teacher need	How Learn4All responds
"What should I study next?"	Seven explainable rules pick the next topic and say why. A mastery gate makes the order of lessons clear.
"I failed a quiz, now what?"	The lesson moves to the top of My Plans with its best score and a direct retake button. A weaker retake never removes earned progress.
"How is my class doing?"	A class heatmap shows every student by topic, with percentages and a colour legend. A monitoring list flags students who need a check-in.
"I read and focus differently."	Preferences for captions, large print, dyslexia-friendly format and screen-reader format, plus dark mode and a Focus Mode with breaks.
"Practice feels like a chore."	XP for each lesson, achievements, streaks, a class leaderboard and study games keep practice rewarding without punishing low scores.
"I forget dates and routines."	A personal calendar with Indian holidays, a weekly timetable and a school routine planner.

Design principle. Every suggestion in the application can explain itself. A student can open How it works and read the exact rules that produced a recommendation, which keeps the system understandable and fair.

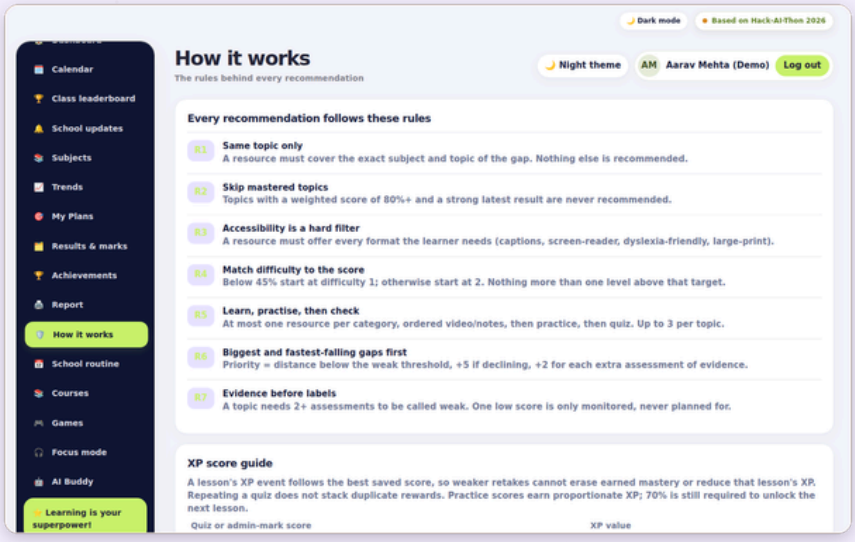


Figure 2. "How it works" lists the rules that choose what to study, so no recommendation is a black box.

TWO WORKSPACES, ONE PLATFORM

3 Users and workflows

Theinterfacechangeswiththerole.Eachpersonseesonly the tools that belong to them.

Student

Registers with a unique ID, name, Indian mobile number, email, class (6 to 10) and password. Then: browses courses, takes lesson quizzes, follows My Plans, checks results, trends and achievements, plans the day, plays study games, uses Focus Mode and the optional AI Buddy.

Teacher / school admin

Signsin with the schooladmin account. Then: reviews the class overview and heatmap, updates marks, publishes targeted notices, monitors students who need follow-up, opens leaderboards for any class and prints student reports.

Access control. The server checks the signed-in role on every protected route. Student accounts that call teacher-only routes receive an HTTP 403 response, and teacher accounts cannot open student-only learning routes. Sessions use HTTP-only cookies and passwords are stored as salted hashes.

Step	Student journey	Teacher journey
1	Create an account or log in with the student ID.	Log in with the school admin ID.
2	Choose a subject and the next unlocked lesson.	Open the class overview and read the heatmap.
3	Answer 15 questions at a chosen difficulty level.	Update marks or review the monitoring list.
4	Score 70% or more to unlock the next lesson, or retake from My Plans.	Publish a notice to everyone, one class or one student.
5	Collect XP, achievements and see the trend.	Print a student report and compare class leaderboards.

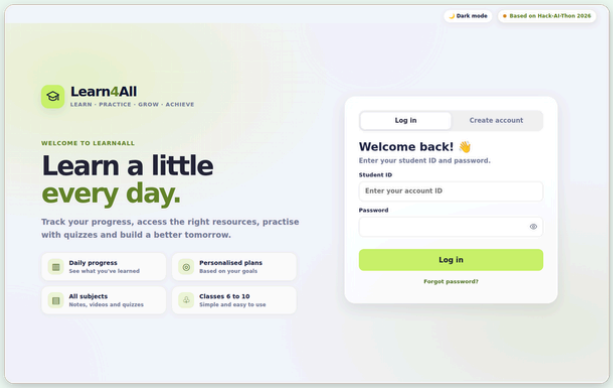


Figure 3. The login and account creation screen.

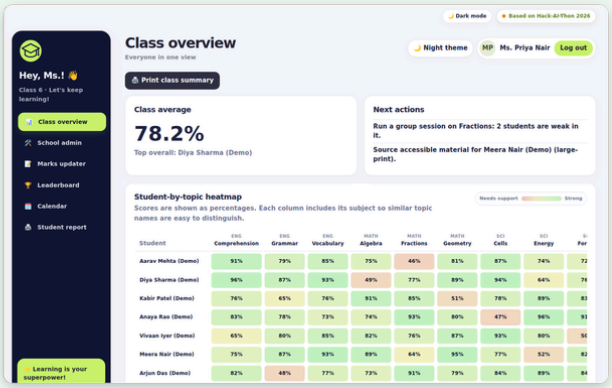


Figure 4. The teacher class overview with the topic heatmap.

THE LEARNING LOOP

4 Course library and quizzes

Onelessonatatime,endinginashortquiz.Passat70%andthenext lesson opens.

The **Courses** section lists lesson titles for the student's class across Hindi, English, Mathematics, History, Politics, Science, Physics, Chemistry and Biology. Physics, Chemistry and Biology are filtered domains within integrated Science, and History and Politics are filters over the combined Social Science books. The catalogue holds **303 lessons** across Classes 6 to 10, with titles and book metadata linked to the NCERT textbook portal. The application stores lesson titles, not copied textbook content.

Class	6	7	8	9	10	Total
Lessons	49	48	60	49	97	303

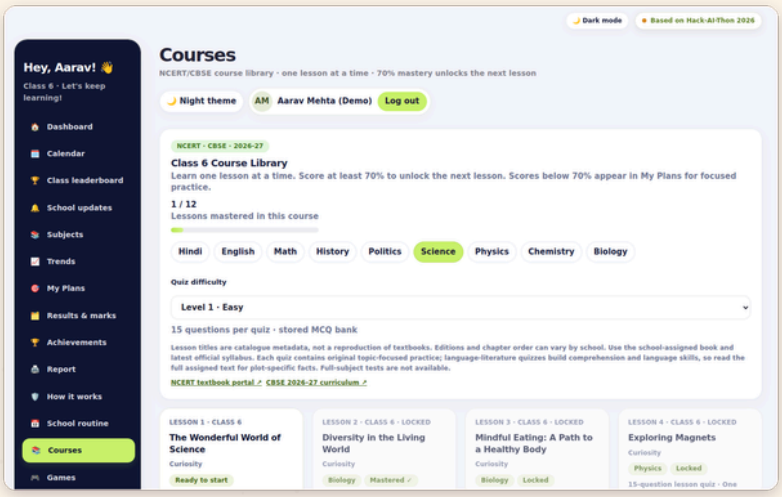


Figure 5. The course library with subject filters, difficulty levels and lesson cards.

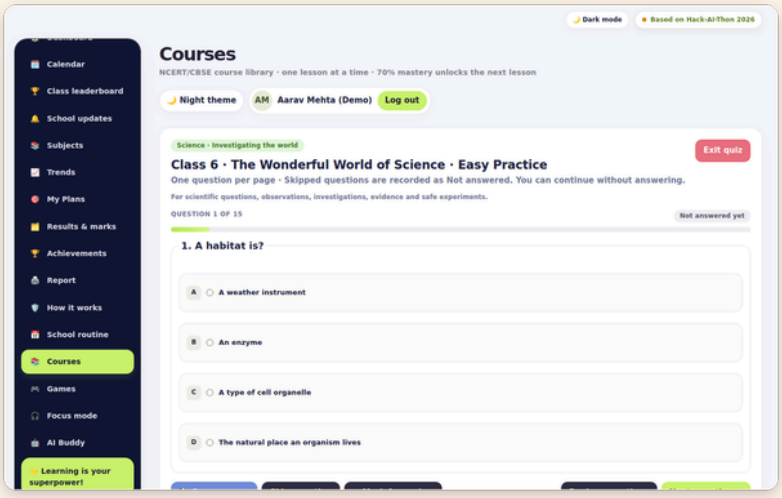


Figure 6. A lesson quiz: 15 multiple-choice questions, answer choices labelled A to D and shuffled on every attempt.

WHAT HAPPENS AFTER THE QUIZ

5 Mastery gate, plans and results

Scores turn into clear next steps, never into dead ends.

- **Quiz review.** After submitting, every question shows all options, the answer the student chose and the correct answer. Skipped questions are counted as unanswered.
- **Three difficulty levels.** Easy, medium and hard question sets let students step up when they are ready.
- **XP that rewards improvement.** XP equals the rounded best score for the lesson, so repeating a quiz cannot farm points and a weaker retake cannot erase progress.
- **My Plans.** Lessons below 70% appear here with their best score, a progress-to-target indicator and a direct retake action. Students can also add their own study plans with a target date.
- **Results and marks.** A history of every attempt and every teacher-entered mark, with scores and dates.

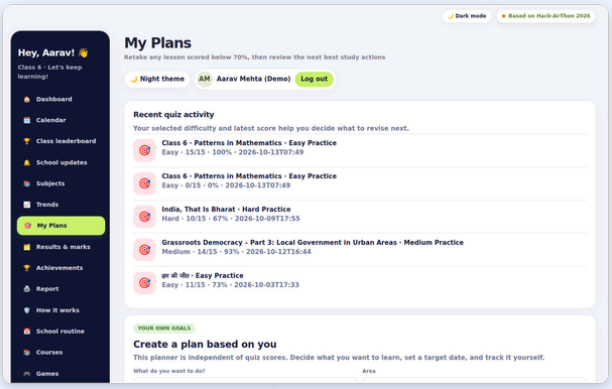


Figure 7. My Plans: recent quiz attempts and a form for personal study plans.

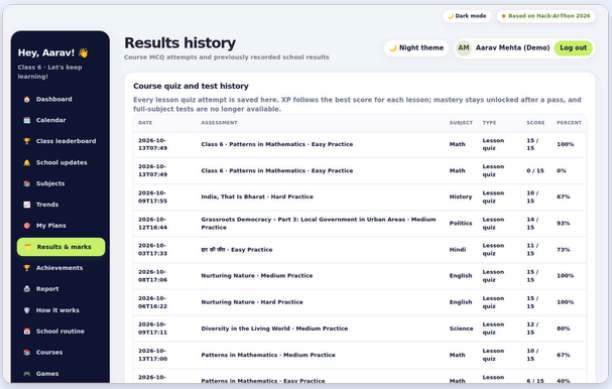


Figure 8. Results history combining quiz attempts and teacher marks.



Figure 9. Subjects and topics with mastery bars and status labels.

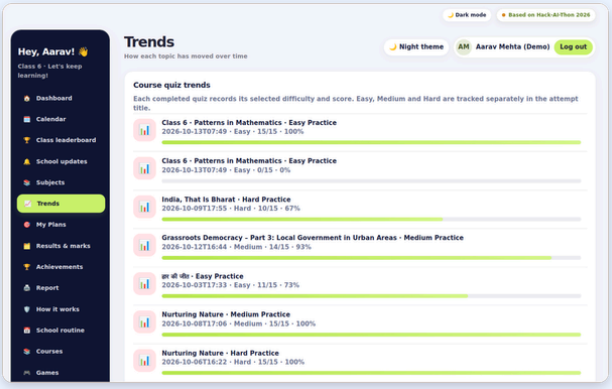


Figure 10. Trends: course quiz scores by lesson over time.

PRACTICE THAT FEELS GOOD

6 Motivation and wellbeing

Achievements, friendly competition and rest are built into the same application.

Achievements track milestones for streaks, results, topic mastery, resources, net XP, games, Daily Wordle, passed lessons and Focus Mode. The **class leaderboard** ranks students by XP within their own class, with marks used only as a tie-breaker. **Study games** (Quick Sprint, Memory Match, Word Scramble and a Daily Wordle) give light practice between lessons.

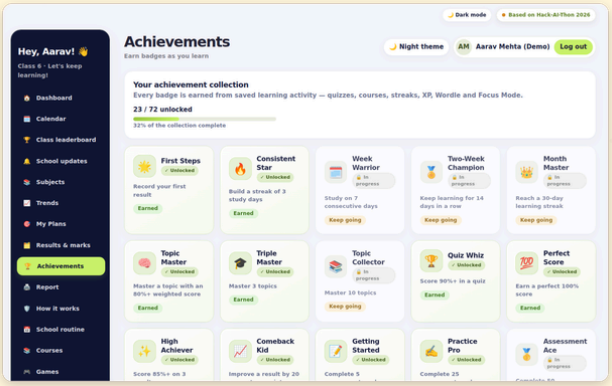


Figure 11. Achievements with progress toward each badge.

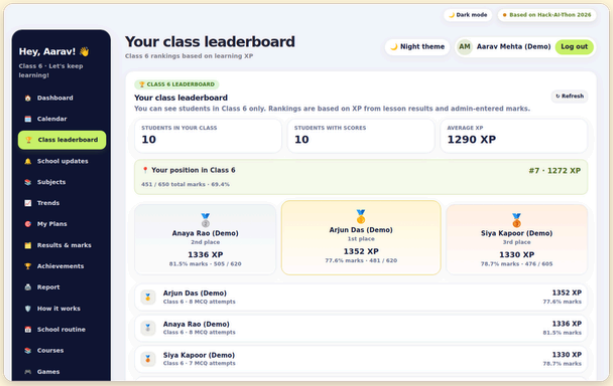


Figure 12. The class leaderboard for a student's own class.

Focus Mode supports study sessions of 5 to 240 minutes. Long sessions are split into 50-minute focus blocks with a 10 or 15 minute break between them. Students get a break reminder and a prompt to drink water about every 30 minutes of focused work. Five bundled soundscapes play from local files, so no third-party audio request is made.

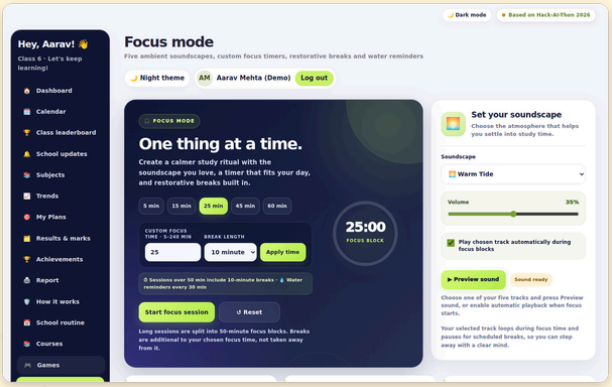


Figure 13. Focus Mode with timer presets, soundscapes and wellbeing reminders.

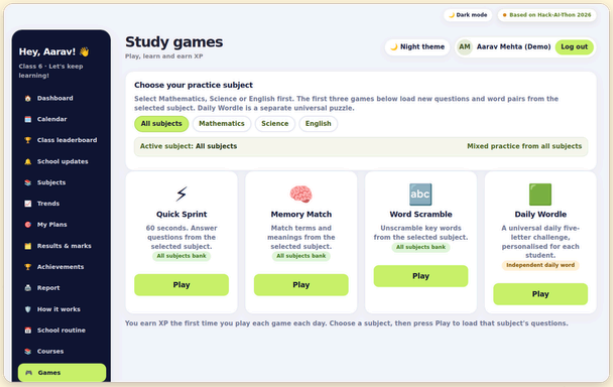


Figure 14. Study games and the Daily Wordle.

NEVER MISS A DATE

7 Planning and school updates

Personal planning and school communication sit beside the learning tools.

- **Calendar.** Each account can add, edit, complete and delete private tasks. The calendar preloads published 2026 Central Government holidays and common national and educational observances, with links to official sources.
- **School routine.** A day-by-day routine with a "today" panel, customizable activities and reminders while the app is open.
- **School updates.** Teachers publish announcements, homework and exam notices and supportive warning notices. Students can mark them as read, and the teacher sees read or pending status per student.
- **Student report.** A printable "progress at a glance" report with subject performance, ready to save as a PDF.

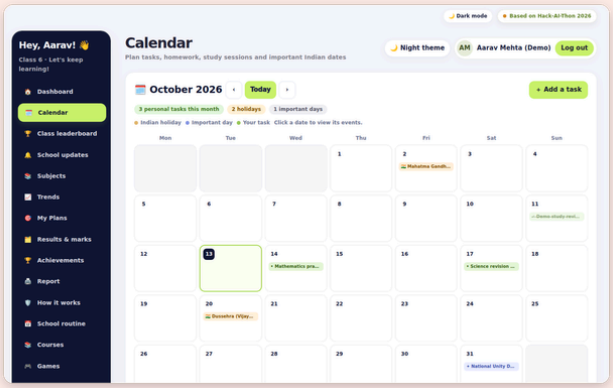


Figure 15. The personal calendar with holidays and tasks.

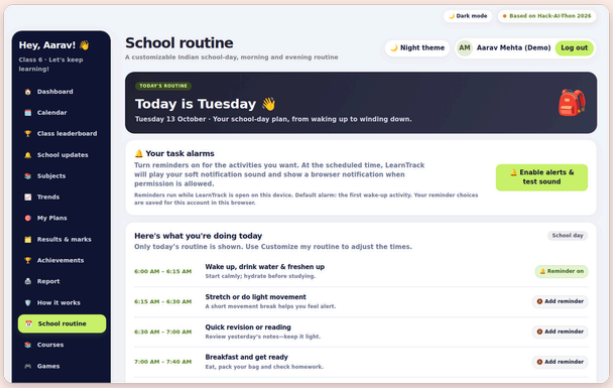


Figure 16. School routine for today.

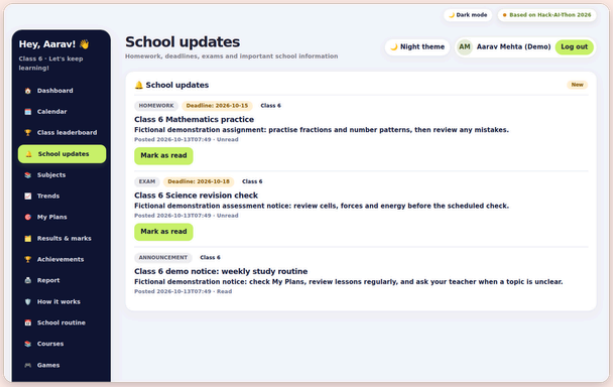


Figure 17. School updates with read receipts.

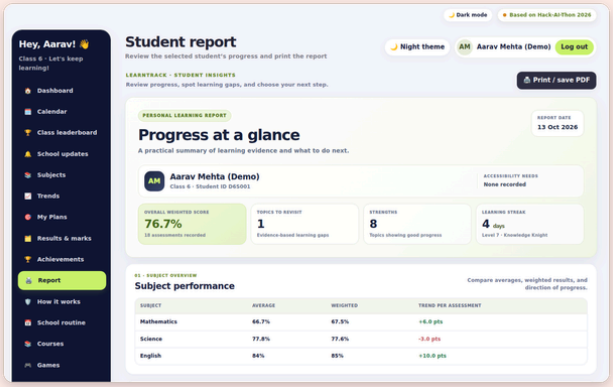


Figure 18. The printable student progress report.

THE WHOLE CLASS AT A GLANCE

8 Teacher and admin tools

Evidenceforthenextconversationwithastudent,withoutreading every answer sheet.

The **Class overview** shows the class average, a next-actions panel and a student-by-topic heatmap with explicit percentages, a colour legend, sticky student names and horizontal scrolling. The **School admin** page publishes notices to everyone, one class or one student and lists a monitoring view of recent activity. The **Marks updater** records teacher-entered marks; saving the same student, title and subject updates the mark rather than adding a duplicate, and the mark also drives the student's XP.

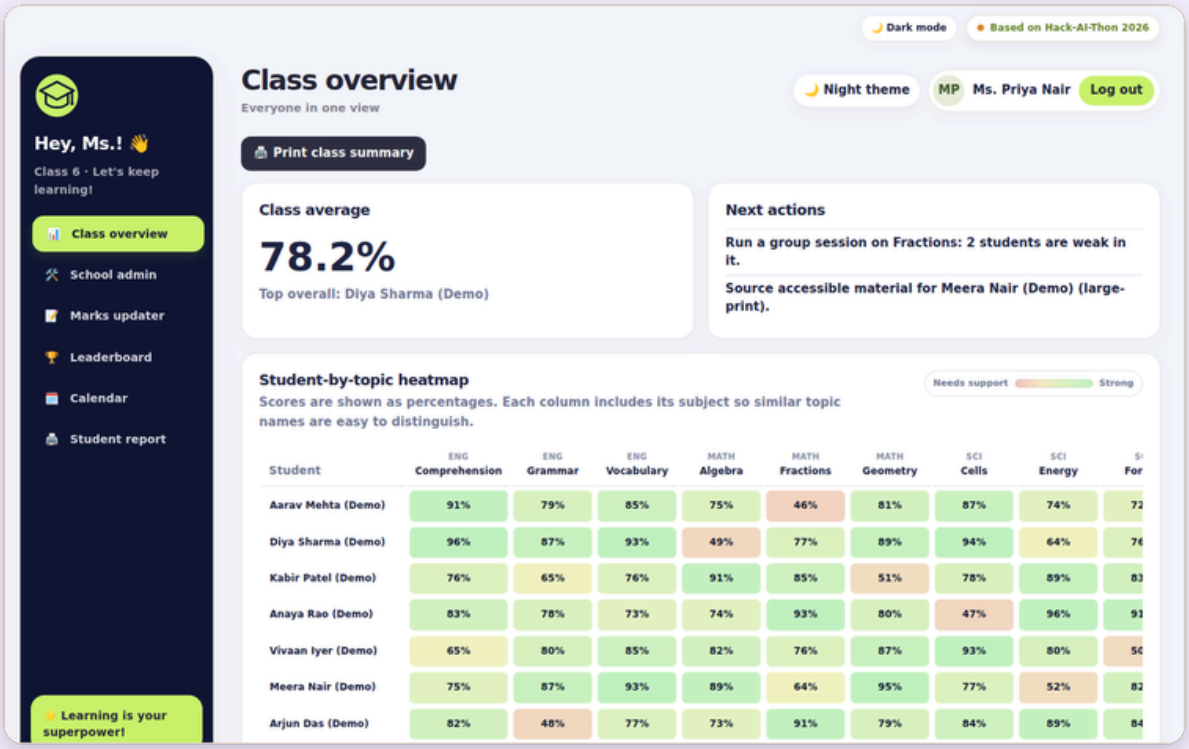


Figure 19. Class overview: class average, recommended actions and the student-by-topic heatmap.

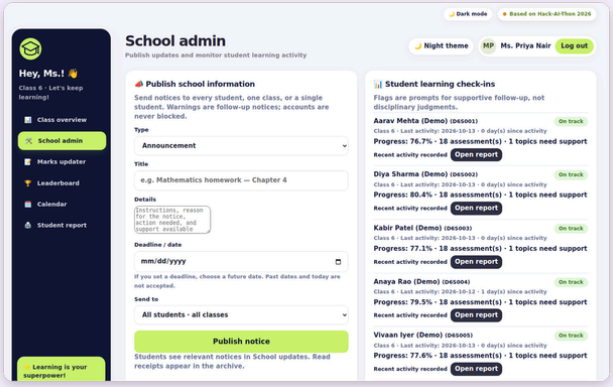


Figure 20. School admin: publish notices and see learner check-ins.

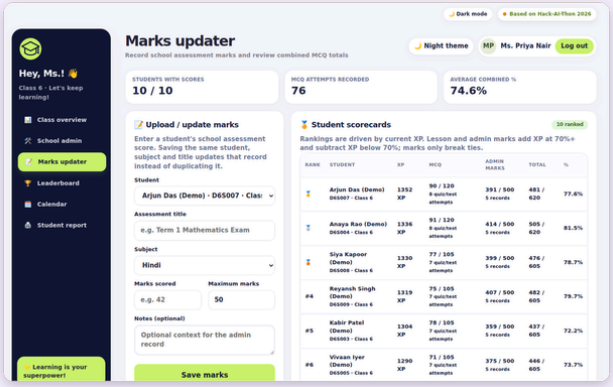


Figure 21. Marks updater with student scorecards.

DESIGNED TO INCLUDE EVERYONE

9 Accessibility and educational impact

Learn4All supportsUNSustainableDevelopmentGoals4and10through

practical, visible features.

Format preferences

Learners canchoose captions,large print, dyslexia-friendly or screen-reader formats, and the app remembers the choice.

Light and dark themes

Athemetoggleoneveryscreen, with high-contrast heatmap text and a labelled colour legend.

Calm pacing

One lesson at atime, three difficulty levels and no penalty XP for low scores keep the pressure humane.

Private and light

Runslocally on plain Python with hashed passwords, so it can work on a modest computer.

Goal	How Learn4All contributes
SDG 4: Quality Education	Structured, syllabus-linked practice for Classes 6 to 10, mastery-based progression, explained recommendations and teacher visibility into progress.
SDG 10: Reduced Inequalities	Choice of learning formats, a local-first design that does not depend on paid services, and a supportive approach to warnings and low scores.

A note on evidence. The scores, streaks and rankings in this document are fictional demonstration data. They show how the application behaves; they are not evidence of learning outcomes. Measuring real impact needs a classroom pilot, which is part of the roadmap.

HOW IT IS BUILT

10 Architecture and technology

A small, readable Python codebase with no third-party dependencies for the core app.

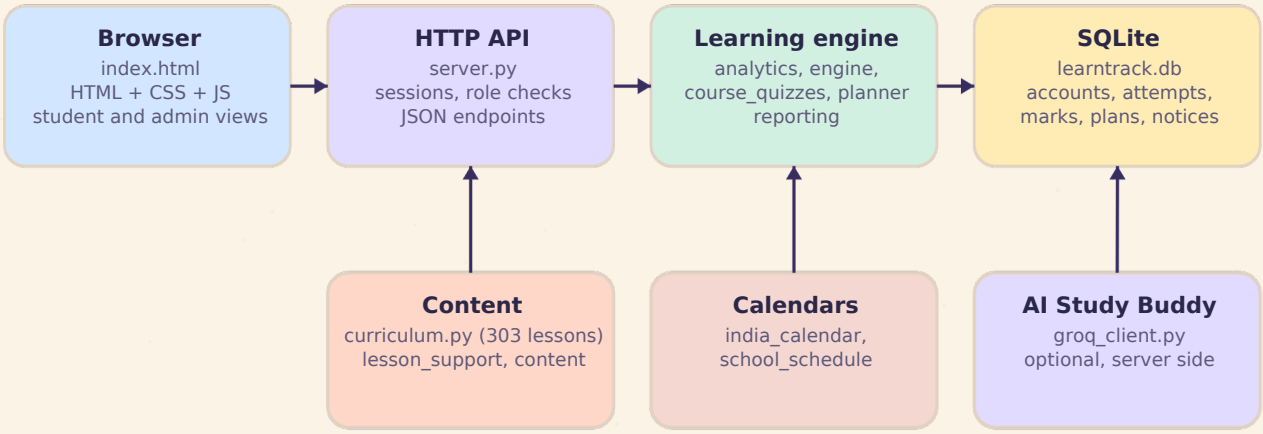


Figure 22. Architecture: the browser talks to a Python HTTP API, which uses the learning engine, content modules and SQLite.

Layer	Technology	Purpose
Language	Python 3.8 or newer	All application logic and the server.
Web server	http.server (standard library)	Serves the page and a JSON API on 127.0.0.1:8765.
Database	sqlite3 (standard library)	Accounts with hashed passwords, attempts, marks, plans, notices, focus sessions.
Interface	HTML, CSS and JavaScript	A single-page interface with student and teacher workspaces, light and dark themes.
AI Study Buddy	Groq chat completions (optional)	Explains topics and quiz answers; the key stays on the server and the app works without it.
Testing	unittest	32 automated tests covering regressions, demo data and the AI client.

Object-oriented structure

models.pydefines Subject, Assessment, Student and a Resource family (Video, Notes, Practice and Quiz subclasses) that share a common interface. **analytics.py**, **engine.py**, **planner.py** and **reporting.py** each own one responsibility: scoring and trends, the seven recommendation rules, the weekly timetable and report building. **validation.py** checks for missing, invalid, inconsistent and duplicate input.

FROM ZIP TO BROWSER IN A MINUTE

11 Installation and running

The only prerequisite is Python 3.

Step	What to do
1	Extract the project ZIP into a new folder and open the **src** folder.
2	Windows: double-click **RUN_LEARNTRACK.bat**. Any system: run python main.py from the src folder.
3	Open http://127.0.0.1:8765 in a browser and keep the terminal window open.
4	Log in with a demo account below. The database is created on first launch with fictional demo data.
5	Optional: copy .env.example to .env and add your own GROQ_API_KEY to switch on the AI Study Buddy.
6	Run the tests with python -m unittest discover -s tests -v. Delete learntrack.db to reset the data.

Demo accounts (local demonstration only)

Role	Login ID	Password	What it shows
School admin	T001	teach123	Class overview, marks, notices, leaderboards
Student (Demo)	D6S001	Demo6!001	Aarav Mehta (Demo), improving Mathematics
Student (Demo)	D6S004	Demo6!004	Anaya Rao (Demo), dyslexia-friendly preference

Ten fictional Class 6 students (D6S001 to D6S010) are preloaded with marks, quiz attempts, streaks, focus sessions,plansand notices. Changethe admin password before any shared deployment.

Suggested live demo

- Login as a student,open Courses,take a lesson quiz and see the mastery result.
- Open My Plans to retake, then check Achievements and the class leaderboard.
- Switch to the admin account to show the heatmap, then publish a notice and watch it arrive for the student.

FROM DEMO TO EVERY CLASSROOM

12 Future scope

Theseare roadmapideas,notpromises.Each step needs real classroom feedback first.

1 Next: production-ready

• RealSMSOTPfor password recovery
• Teacher-reviewed question banks
• Live testing of the AI Buddy
• Forced admin password change on first login

2 Soon: more learners

• Installable mobileapp
• Hindi and regional languages
• Parent view and weekly digest
• Classes 1 to 5 and 11 to 12

3 Later: smarter and wider

• Adaptivedifficultyfromreal data
• School and district dashboards
• Offline-first sync
• Teacher-made lessons

Known limits of the current version

- Thequestionbanks are originalpracticequestionsandneedteacher review for chapter-level accuracy; they are not the NCERT question bank.
- Password recovery shows the one-time code on screen for demonstration. No SMS is sent.
- Class 10 language tracks and syllabus details can vary by school, so the app marks them for teacher confirmation.
- The calendar contains published 2026 central holidays; state and school holidays may differ.
- The AI Study Buddy needs an internet connection and the user's own Groq key; the rest of the app works offline.

THE PEOPLE AND SOURCES BEHIND IT

13 Team and references

Creditsfollow theprojectpresentation.

Aaditya Sorout
Developer.Wrotethecodebase together with Syed Sarfaraz Ahamed.

Syed Sarfaraz Ahamed
Developer.Wrotethecodebasetogether with Aaditya Sorout.

Pramveer Kumar
Foundation codeand ideas.

Sarah
Research.

With thanks: Mohamed Aathif (team member), Mohamed Rayan M (team member), Sriyansh Sai Mohanty (team member) and M. Vetri Selvan M (team member).

References

[1] NCERT textbook portal, Classes I to XII. https://ncert.nic.in/textbook.php

[2] CBSE Academic, curriculum 2026-27. https://cbseacademic.nic.in/curriculum_2027.html

[3] India Post, Holidays list. https://www.indiapost.gov.in/holidays-list

[4] National Portal of India, Holiday calendar. https://www.india.gov.in/calendar

[5] Government of India, Central Government holiday list. https://cgca.gov.in/ccadl/list-of-holiday

[6] United Nations, Sustainable Development Goals 4 and 10. https://sdgs.un.org/goals

[7] Python documentation: http.server, sqlite3, unittest. https://docs.python.org/3/

[8] Groq Console (optional AI Study Buddy). https://console.groq.com/

External links were not re-checked while preparing this document. Lesson titles and calendar dates should be confirmed against the official sources before classroom use.

Learn4All

Myra Global Tech HACK-AI-THON 2026

All learner data shown in this document is fictional demonstration data.